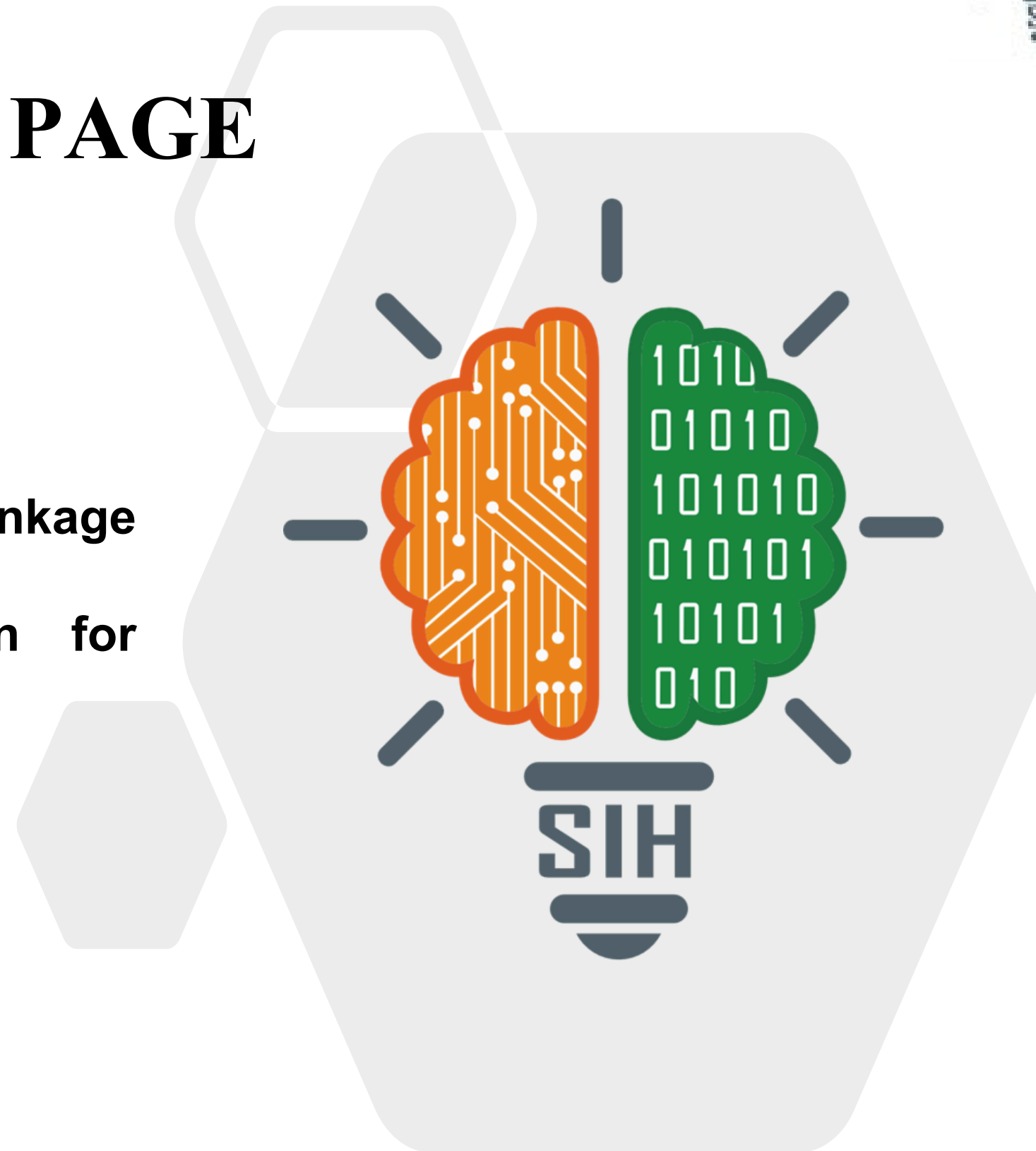
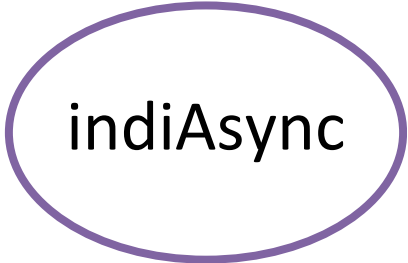


TITLE PAGE

- Problem Statement ID – SIH26090
- Problem Statement Title- AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans
- Theme- Heritage & Culture
- PS Category- Software
- Team ID-
- Team Name (Registered on portal) - indiAsync





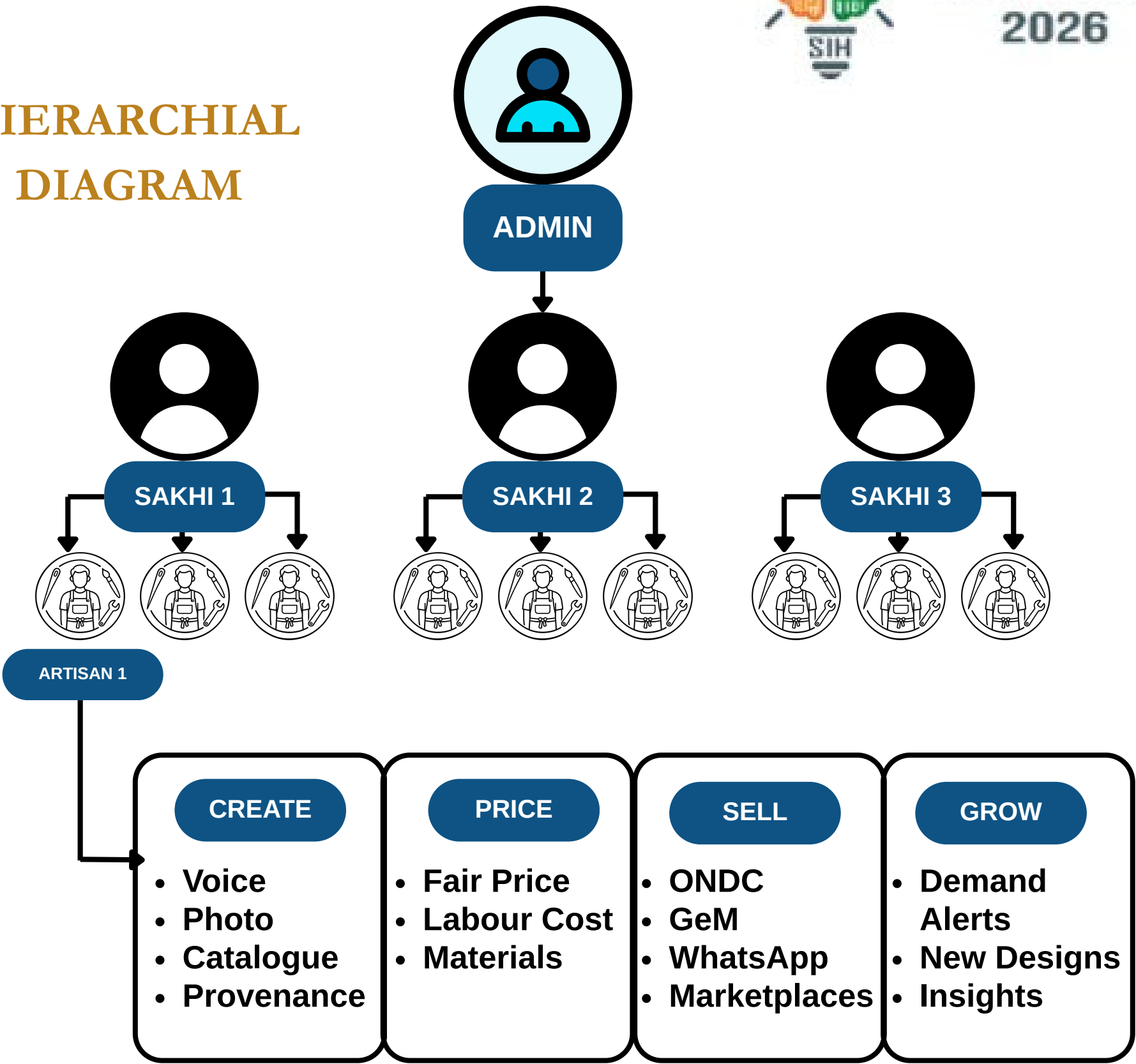
HOW IT WORKS

- Artisan speaks in her language, points her camera at her product - the app does the rest.
- AI removes backgrounds, corrects lighting, and creates professional product photos on-device.
- Voice note to SEO-ready listing in English & Hindi, using only confirmed facts.
- ML-based pricing suggests a fair price while protecting her labour with a minimum-wage floor.
- One 60-second stock video creates bulk listings for 40 products, with her narration attached.
- One upload publishes to ONDC, GeM & marketplace-ready formats, reaching buyers across channels.
- WhatsApp Agent answers buyer questions 24/7 using her past replies, escalating sensitive queries to her.
- KalaSetu learns from product performance, showing her what is selling, what needs improvement, and what to make next.

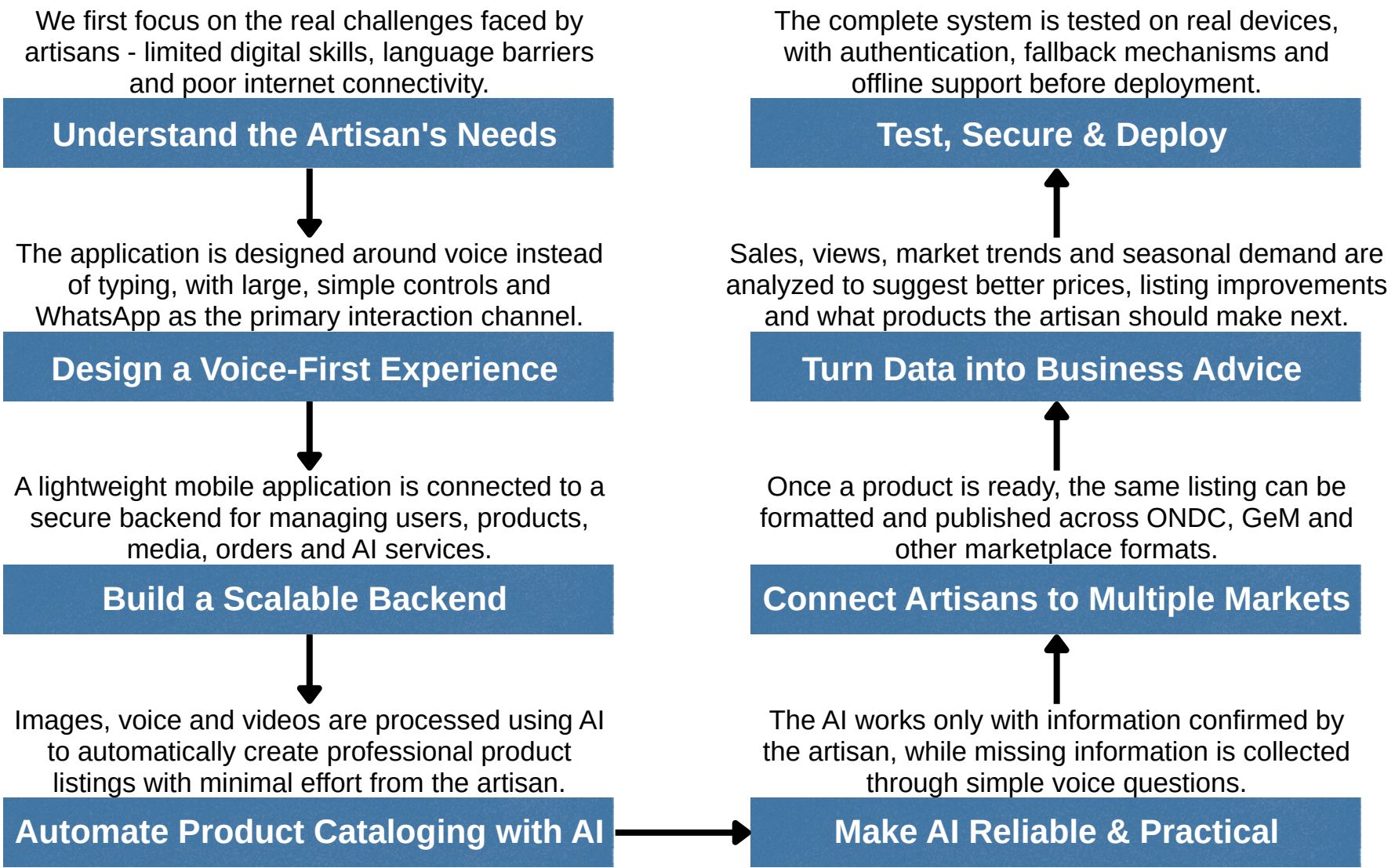
HOW IT ADDRESSES THE PROBLEM

- Multilingual voice, read-aloud UI and offline processing eliminate typing and reduce dependence on internet connectivity.
- AI turns photos, voice and videos into enhanced, grounded listings with bulk cataloguing and automatic craft/GI identification.
- ML combines market, material and labour signals while a dignity floor ensures her labour is never undervalued.
- Publish to ONDC, GeM and marketplaces, while WhatsApp RAG provides continuous, artisan-specific buyer support.
- Demand signals, stale-listing coaching and motif mockups help artisans decide what to make, improve and sell next.
- Sakhi mode, credit reports and 3D previews extend KalaSetu from individual artisans to scalable, richer digital commerce.

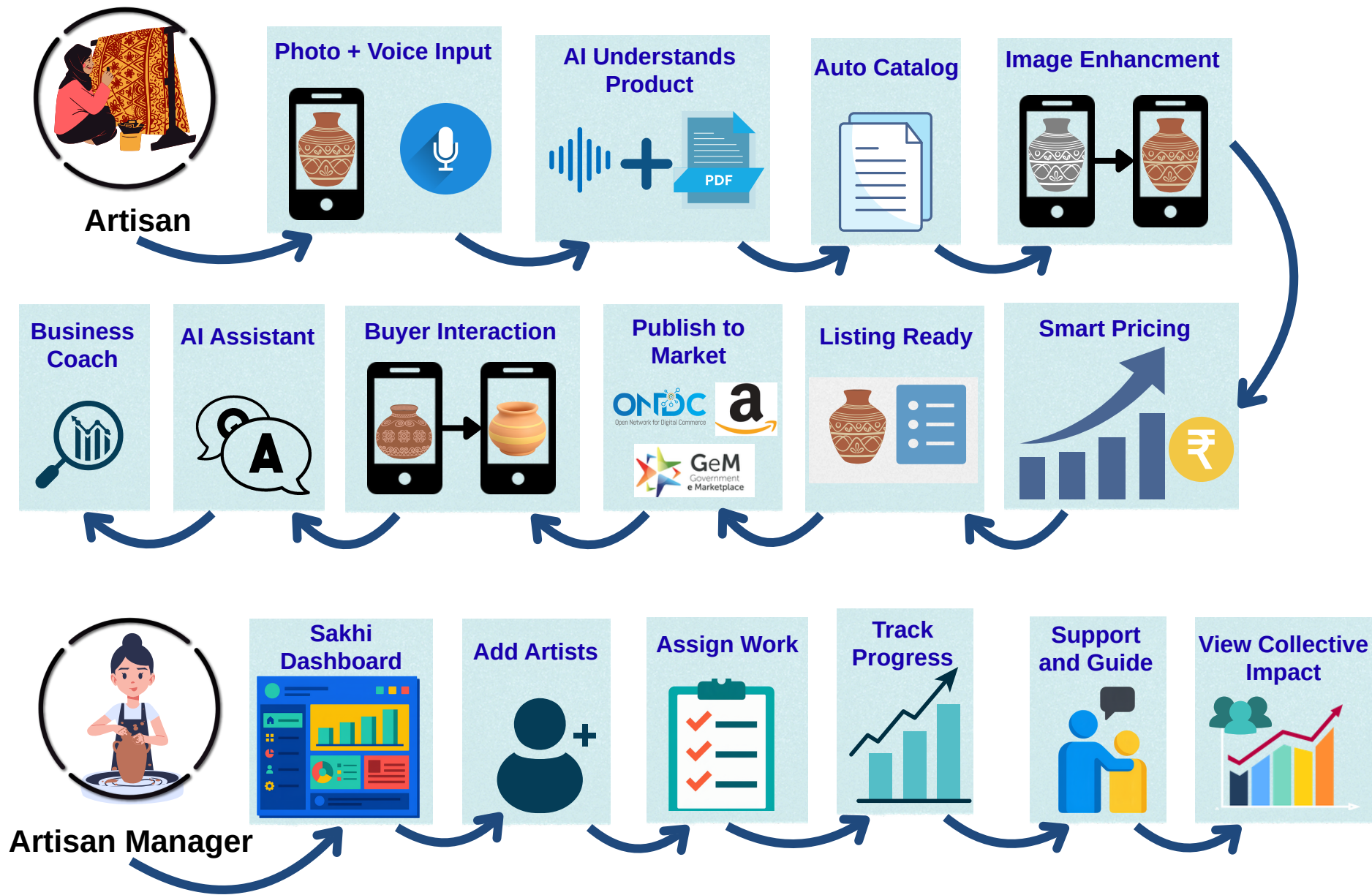
HIERARCHIAL DIAGRAM



METHODOLOGY



FLOW CHART



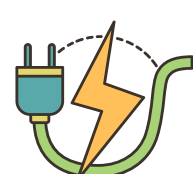
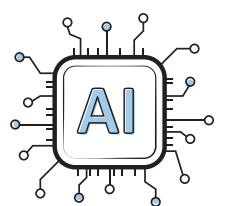
TECH STACK



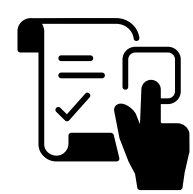
ANALYSIS OF THE FEASIBILITY OF THE IDEA

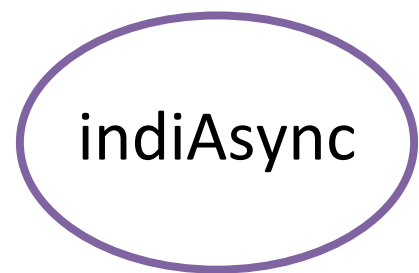
1. TECHNICAL FEASIBILITY Built on a proven, entirely free-tier stack (FastAPI, Neon Postgres with pgvector, Redis, Expo) with a fail-soft AI pipeline — Gemini/Groq LLM fallback and cloud/local speech fallback — so no single provider outage ever blocks a listing.	2. ECONOMICAL FEASIBILITY Zero infrastructure cost to pilot on free tiers, and no new marketplace to build or market — the app publishes directly into ONDC, GeM and existing e-commerce channels , riding their buyer base instead of acquiring one.	3. SOCIAL FEASIBILITY A zero-text, voice-first interface in regional languages reduces dependency on physical fairs and raises digital literacy , with a "graduation" design that lets AI assistance fade as the artisan's own skills grow.	4. LEGAL & COMPLIANCE FEASIBILITY The description engine generates only from artisan-confirmed facts , marking anything unverified as "not specified" rather than guessing, while GI-tag and provenance fields keep every listing compliant with Geographical Indication norms.	5. OPERATIONAL FEASIBILITY A WhatsApp-first , app-second model lets an artisan list by voice note and photo without opening the app, while one listing auto-formats across ONDC, GeM and marketplace CSVs — inventory managed at a place.	6. SECURITY FEASIBILITY Phone + OTP + PIN authentication with JWT sessions and rate-limited logins, backed by tenant isolation enforced at the database query layer on every endpoint, so one artisan's data can never be read by another.
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POTENTIAL CHALLENGES AND RISKS

LOW DIGITAL LITERACY & TRUST Artisans may distrust an unfamiliar app after past scheme fatigue and prior failed digital rollouts. 	CONNECTIVITY & POWER RELIABILITY Rural 2G/3G coverage and intermittent electricity interrupt uploads and live AI calls. 
AI ACCURACY & HALLUCINATION RISK Misclassified crafts or invented product claims would erode buyer and artisan trust. 	CHANNEL DEPENDENCY Reliance on ONDC, GeM and WhatsApp APIs that sit outside our control and can change. 

STRATEGIES FOR OVERCOMING THESE CHALLENGES

VOICE-FIRST + SAKHI-ASSISTED ONBOARDING A trusted local coordinator bridges the first-use gap, so adoption never starts from a cold install. 	OFFLINE-FIRST, LOW-BANDWIDTH DESIGN Compressed uploads, local speech fallback and queued sync — nothing blocks on a live connection. 
GROUNDING GENERATION + HUMAN CONFIRMATION The AI states only artisan-confirmed facts, and she reviews every listing before it is published. 	PUBLISHER ARCHITECTURE, NOT ONE MARKETPLACE One listing feeds many channels, so a single API change never stops sales anywhere else. 



IMPACT AND BENEFITS



TARGET AUDIENCE

- **Artisans & Weavers** – year-round digital sales without literacy or technical skill
- **SHG Women & Sakhi Leaders** – one login manages 15–20 artisan accounts across a village
- **Scheme Beneficiaries** – PMEGP, AHVY and MUDRA recipients converting grants into sustained income
- **B2B & Export Buyers** – verified, GI-tagged supply sourced directly from the maker
- **Government e-Marketplaces** – ONDC and GeM gain authenticated artisan sellers at scale
- **Ministries & NGOs** – live visibility into cluster performance and scheme outcomes

IMPACTS

Year-Round Income



Replaces the 8–10 fair days a year with a 365-day presence on ONDC, GeM and marketplaces — **35× more selling days**.

Zero-Literacy Access



Voice-only operation in Indian languages cuts onboarding from weeks of training to **6 minutes**. No reading, no typing, anywhere.

Middleman Elimination



One voice note publishes to ONDC, GeM, Amazon and WhatsApp, letting artisans keep **40–60%** of the final price instead of a trader's margin.

Scale Without New Infrastructure



One trained Sakhi onboards 15–20 artisans, turning the existing SHG network into the delivery channel — national reach with **no new field staff or budget line**.

Make What Sells



Festival and demand signals tell her what to produce before she produces it, **ending the unsold inventory** that piles up after every fair.

BENEFITS

SOCIAL

Sakhi mode lets one SHG leader onboard and manage 15–20 artisan accounts with **earnings kept separate**, so payments reach each artisan own account

ECONOMIC

AI motif mockups place her existing design on tote bags, mugs and laptop sleeves, moving a **₹250 craft item into a ₹1,200** lifestyle product

CULTURAL

A CLIP-based classifier identifies the craft and auto-attaches its **GI tag** and heritage note to every listing, where documented provenance earns a **2–3× premium**

MARKET ACCESS

One **voice note** publishes to **ONDC, GeM, Amazon and WhatsApp** simultaneously, giving a village artisan the **same distribution reach as an organised exporter**

ENVIRONMENTAL

Demand-based production cuts overstock waste. Khadi, jute and natural-dye crafts are auto-tagged sustainable, linking eco-conscious buyers to **zero-emission village production**.

GOVERNMENT PORTALS & STANDARDS

- **Office of DC (Handicrafts), Ministry of Textiles** — handicrafts.nic.in — artisan cluster data, scheme structure and Pehchan card; defines our target beneficiary
- **ONDC — Open Network for Digital Commerce** — ondc.org — primary publishing channel; listings pushed via signed Beckn API calls
- **GeM — Government e-Marketplace** — gem.gov.in — public procurement channel; catalogue exported in GeM-compatible format
- **4th All India Handloom Census (2019–20)** — texmin.nic.in — sector size, artisan income baseline and migration trends
- **MUDRA Yojana** — mudra.org.in — loan eligibility criteria drive the auto-generated income statement feature

RESEARCH & TECHNICAL FOUNDATIONS

- **Vision** — CLIP, IS-Net, InstantMesh — craft classification, background removal, single-image 3D reconstruction
- **Language** — Whisper, multilingual-e5, BGE-Reranker — speech-to-text, cross-lingual embeddings, RAG reranking
- **Pricing** — LightGBM quantile regression, SHAP — price band prediction with human-readable explanations
- **Retrieval** — RAG, Reciprocal Rank Fusion, pgvector HNSW — hybrid dense + sparse buyer query answering
- **Market data** — iTokri, Okhai, Gaatha, Amazon Karigar, IndiaMART — ~5,000-listing price corpus for training and B2B benchmarks